

Research Update: OCR-Derived Egyptian Medicine Search

Meeting 10 through 19 July 2026

Generated 19 July 2026

Abstract

This report consolidates the work completed from Meeting 10 through the latest Algorithm 4 enhancement. It joins 595 OCR observations from 14 source models, 464 collision-free unique OCR query-target pairs, 10 retrieval systems, 24 Algorithm 4 configurations, eight equal-distance policies, and a deterministic 115,000-case synthetic benchmark. Algorithm 4 reaches 47.6293% Hit@1 and 71.1207% Hit@20 on the primary OCR denominator. It exceeds exhaustive Levenshtein by 13.3621 percentage points at Hit@1 and Jaro-Winkler by 3.6638 points at Hit@20. Compact edit distance four or more creates 120 of 134 primary top-20 misses. Removing family rescue lowers Hit@20 by 20.0431 points. On the full synthetic benchmark, Algorithm 4 reaches 82.1165% Hit@1, 93.4122% Hit@20, 93.6374% behavior success, and zero unsafe confident first results. The pharmacist study has 1,125 balanced assignments but no participant outcomes.

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1 Key Findings

- **Algorithm 4 has the best tested OCR retrieval result: 47.6293% Hit@1 and 71.1207% Hit@20.** On the same 464 unique fair pairs, the strongest classical alternatives reach 34.2672% Hit@1 and 67.4569% Hit@20, so the gain is mostly first-rank ordering plus a smaller candidate-recall gain.
- **Four compact edits mark the practical retrieval break.** Hit@20 falls from 93.5484% for two-to-three edits to 41.9355% for four-to-five edits and 4.3478% for six or more, which concentrates 89.55% of primary top-20 misses at distance four or above.
- **Family rescue is the dominant Algorithm 4 component.** Removing it reduces Hit@1 from 47.6293% to 27.3707% and Hit@20 from 71.1207% to 51.0776%, with 96 gains versus two losses for complete Algorithm 4 at rank 1.
- **Safety cannot be inferred from retrieval accuracy.** Removing the clarification gate leaves Hit@1 and Hit@20 unchanged but raises unsafe confident first results from 0% to 24.7845%.
- **Algorithm 4 improves Algorithm 3 on the 115,000 paired cases.** Hit@1 rises by 1.0817 points and Hit@20 by 0.3174 points, with zero unsafe confident first results for both systems.
- **Human-study claims are not available.** The counterbalanced protocol contains 15 participants, 75 crops, and 1,125 assignments, but accuracy, decision time, and safety cannot be reported until real responses and adjudicated labels exist.

2 Experiment Log

The log separates completed measurements from prepared work so design artifacts cannot be mistaken for experimental outcomes.

3 Data and Evaluation Contract

This section fixes the inputs, units, labels, and denominators before comparing systems.

Experiment	Evaluation unit	Cases	Primary output	Status
OCR source audit	One model prediction for one verified family	595	Error severity, operation mix, evidence, source-model case mix	Complete
Primary OCR search	One collision-free unique compact query-target pair	464	Hit@1, Hit@20, MRR@20, safety, latency	Complete
Retrieval comparison	Same primary pair under each system	4,640	Ten-system metrics and paired effects	Complete
Algorithm 4 ablation	Same primary pair under each configuration	11,136	Complete system plus 23 one-component removals	Complete
Equal-distance policy test	Same primary pair under each policy	3,712	Eight all/development/holdout comparisons	Complete
Synthetic benchmark	One deterministic generated search case	115,000	Four algorithms, 34 categories, six mistake mechanisms	Complete
Enhancement regression	Locked manual, OCR, browser, sample, and full suites	122,109	Before/after retrieval and paired switches where stored	Complete
Pharmacist study	Participant-case-condition decision	1,125 assigned	No tool, DrugEye, Algorithm 4	Prepared, not executed

Table 1: All source tables lack run timestamps. “Meeting 10 through 19 July 2026” therefore denotes the artifact lineage and regeneration date, not a row-level time interval.

Stage	Concrete example	Meaning
Input	OCR text MYMLAX	Text already produced by an OCR model; no image enters this search benchmark.
Process	Search 25,066 catalog medicines	Generate candidates, group commercial variants by family, rank evidence, and apply clarification gates.
Output	MYOLAX at rank 1	Ranked commercial family plus response state and evidence.
Hidden label	Verified family MYOLAX	Used only after retrieval to calculate metrics.
Score	Hit@1 = 1; Hit@20 = 1	The verified family appears first and therefore also within the first 20.

Table 2: The verified target never enters candidate generation or ranking. The benchmark measures search recovery after OCR, not prescription segmentation or OCR recognition accuracy.

Human-readable source	Rows	Fields	Role
OCR predictions	595	15	OCR text, verified target, source alignment, and model identity.
OCR search cases	595	43	Fair-score labels, evidence counts, severity cohorts, and split.
Latest OCR Algorithm 4 results	595	44	Ranked families, decision state, safety, and latency.
Retrieval and ablation case results	20,230	29	Ten retrieval systems and 24 Algorithm 4 configurations.
Retrieval aggregate metrics	612	18	Metrics by denominator, split, cohort, distance, and mistake type.
Retrieval paired comparisons	32	12	Exact gains, losses, and McNemar tests.
Synthetic test cases	115,000	18	Deterministic generated inputs with category and provenance.
Synthetic Algorithms 1–4	460,000	29–39	Four aligned row-level outcome tables.
Synthetic category metrics	156	16	Four algorithms over all scopes and 34 categories.
Equal-distance evidence	595	27	Candidate evidence and eight policy outcomes.
Pharmacist assignments	1,125	5	Prepared participant-case-condition schedule, no responses.

Table 3: Every source file was read in full. One-to-one keys reconcile the OCR tables, and all four synthetic result tables share the same 115,000 source rows.

3.1 Input, Process, Output

3.2 Source Inventory

3.3 Data Dictionary

Raw names appear here once; later sections use human labels. The generated companion dictionary contains 356 dataset-field entries, including fields not needed in the report body.

Compact form lowercases text and removes non-alphanumeric characters. For **CLEX A NE** → **CLEXANE**, source alignment counts two spaces as edits, while both compact strings are **cllexane** and compact distance is zero. With compact query q_c , target y_c , and Levenshtein distance L ,

$$d = L(q_c, y_c), \quad r = \frac{d}{\max(|y_c|, 1)}.$$

An extreme prediction has $r > 0.60$. For **GHONT** → **CALAMINE**, $d = 7$, $|y_c| = 8$, and $r = 0.875$. The 0.60 threshold is an operational review band, not a probability or clinical-risk estimate.

3.4 Denominators

3.5 Metrics and Statistical Tests

For query q , $R_k(q)$ is the first k returned families and $Y(q)$ is the verified relevant-family set:

$$\text{Hit@k}(q) = \mathbf{1}[R_k(q) \cap Y(q) \neq \emptyset].$$

- Hit@1 failure with Hit@20 success is a ranking failure; Hit@20 failure is a candidate-retrieval failure.

Raw field	Human meaning
edited_name	OCR-predicted medicine text supplied to search.
matched_canonical_name_norm	Human-verified commercial medicine family.
edit_distance_count	Source alignment edits, which can retain spaces.
edit_distance	Compact Levenshtein edits after case, punctuation, and spaces are normalized.
normalized_edit_distance	Compact edit distance divided by compact target length.
shared_ngram_count	Count of adjacent-character pairs shared by compact query and target.
analysis_cohort	Mutually exclusive severity or ambiguity cohort.
scored_case	Whether a row belongs to the fair single-answer denominator.
hit_at_1, hit_at_20	Verified-family recovery by rank 1 or rank 20.
unsafe_confident_top1	Wrong first family presented confidently without clarification.
latency_ms	Warm query time in milliseconds, excluding index preparation.
mistake_type	Search failure mechanism, independent of mutation category.

Table 4: The complete generated dictionary prevents raw implementation names from leaking into figure titles, axes, legends, or analytical claims.

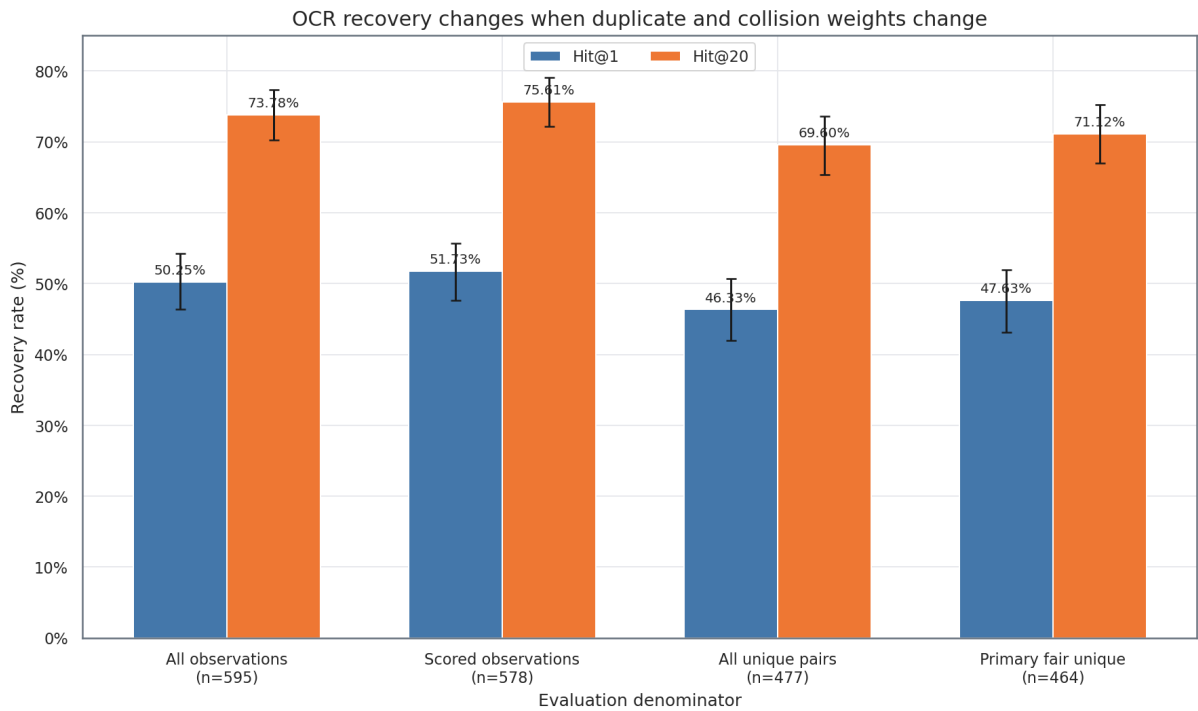


Figure 1: Each row is one denominator; the x-axis reports Hit@1 and Hit@20. Inclusive observations count every OCR-model output, scored observations remove 17 real-name collisions, unique pairs remove repeated compact query-target pairs, and the primary denominator applies both rules.

Insights

1. The number to cite for search comparison is 47.6293% Hit@1 and 71.1207% Hit@20 over 464 primary pairs. (Each distinct fair query-target pair receives one vote.)
2. The inclusive 595-row result is higher, 50.2521% Hit@1 and 73.7815% Hit@20. (Repeated OCR outputs receive repeated votes, so it answers a data-audit question rather than a pair-balanced comparison.)
3. A collision such as RIVOTRIL → RIVO stays in diagnostics but not fair single-answer accuracy. (The entered text is already another exact catalog family.)

- MRR@20 uses reciprocal rank for the first relevant result through rank 20 and zero beyond rank 20.
- Paired risk differences use 10,000 bootstrap resamples with seed 20260719. Error bars are 95% percentile intervals.
- Exact two-sided McNemar tests use pairwise gains and losses. Matched odds ratios describe effect direction. Exploratory p-values are not adjusted for multiple comparisons and are not treated as confirmatory evidence.
- Spearman correlation measures monotonic association. It does not assign causality to edit distance, evidence counts, or recovery.

4 OCR Dataset Composition

The 595 observations mix unequal model coverage, target-family-disjoint splits, and error severities, so raw model percentages require case-mix context.

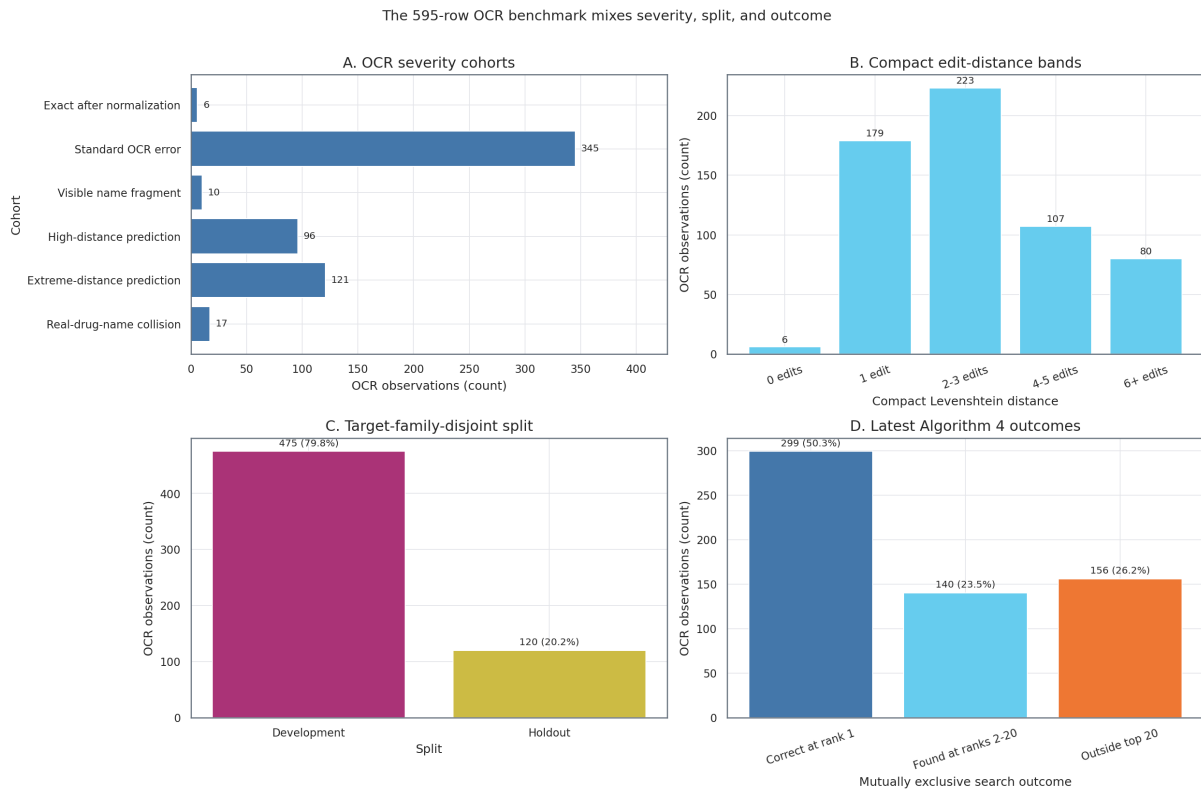


Figure 2: Panel A counts severity cohorts, panel B counts compact edit-distance bands, panel C counts development and holdout observations, and panel D partitions latest Algorithm 4 outcomes. Every panel uses the same 595-row inclusive denominator.

Insights

1. Extreme predictions account for 121 rows, or 20.34%, while standard OCR errors account for 345, or 57.98%. (The dataset is not dominated by ordinary one-edit spelling errors.)
2. Algorithm 4 places 299 observations at rank 1, 140 at ranks 2–20, and leaves 156 outside the top 20. (The middle group needs reranking; the last group needs broader candidate generation.)
3. The split contains 475 development and 120 holdout observations, with zero target-family overlap. (All KETOROLAC rows are development; all VIGOREX rows are holdout.)

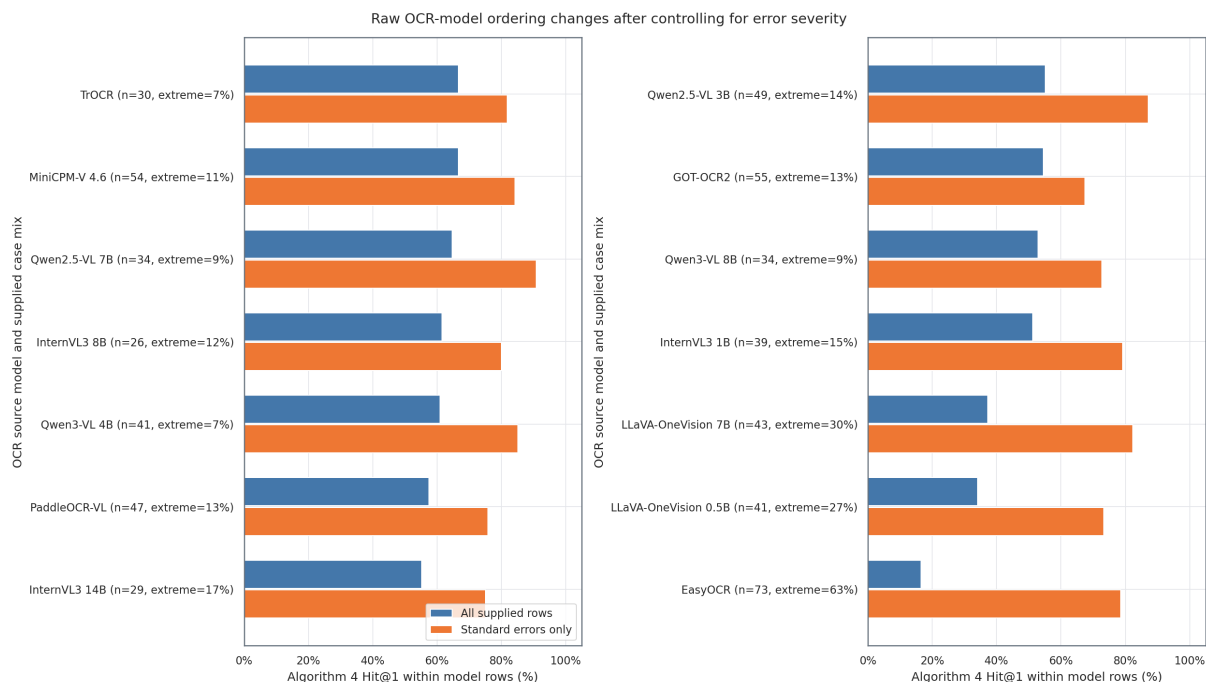


Figure 3: For each OCR source model, blue is Algorithm 4 Hit@1 on all supplied rows and orange is Hit@1 on standard errors only. Labels show model row count and its extreme-row share; models did not receive a common image set.

Insights

1. Raw downstream search rates are not an OCR leaderboard. (The 14 models contribute 26–73 observations with different targets and corruption.)
2. EasyOCR rises from 16.44% raw Hit@1 to 78.57% on its 14 standard rows. (Forty-six of its 73 supplied rows are extreme, so severity explains a large part of the raw score.)
3. A fair OCR comparison requires every model to process the same crops with the same preprocessing and target labels. (This dataset supports search analysis, not matched OCR-model ranking.)

5 OCR Error Severity and Evidence

This section tests which measurable error properties are associated with candidate recovery on the 464 primary pairs.

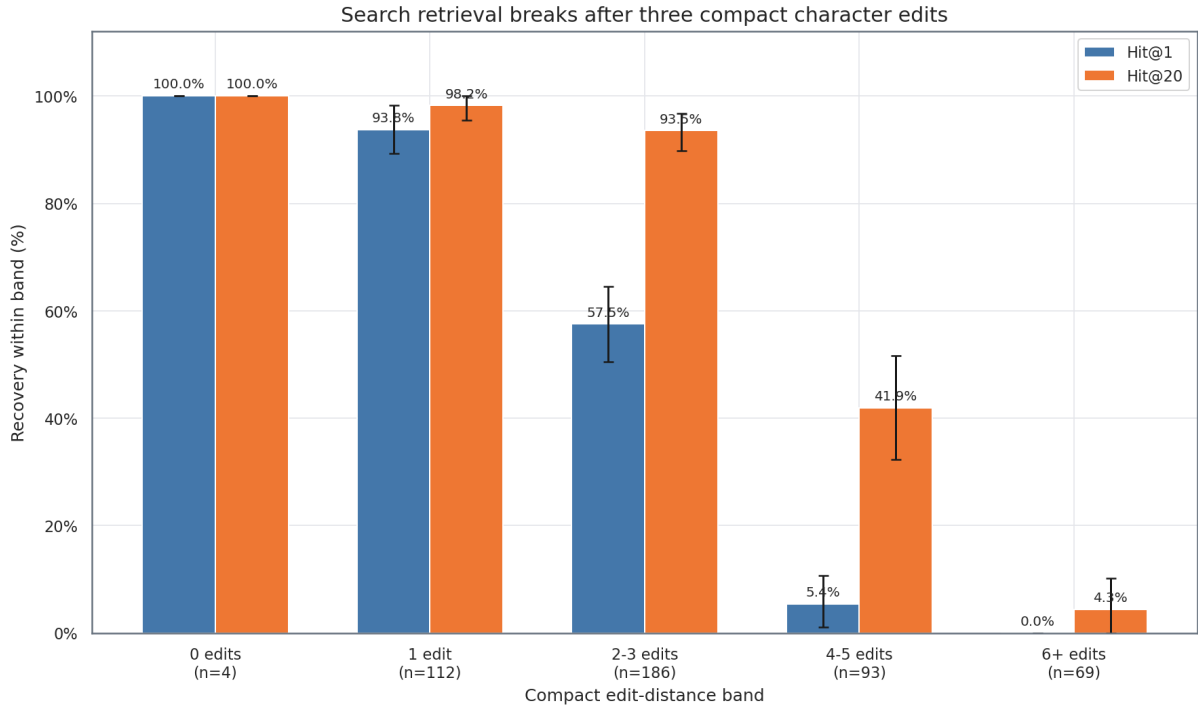


Figure 4: The x-axis groups compact Levenshtein edit counts; blue is Hit@1 and orange is Hit@20 within each group. Error bars are 95% bootstrap intervals and labels print the group denominator.

Insights

1. One-edit queries remain recoverable at 93.75% Hit@1 and 98.21% Hit@20. (The catalog and current signals handle ordinary OCR corruption.)
2. Four-to-five edits fall to 5.38% Hit@1 and 41.94% Hit@20; six or more fall to 0% and 4.35%. (Distance four is the observed retrieval break.)
3. Distance four or above creates 120 of 134 primary top-20 misses, 89.55%. (Work on severe candidate generation has greater reach than tuning already-correct one-edit cases.)

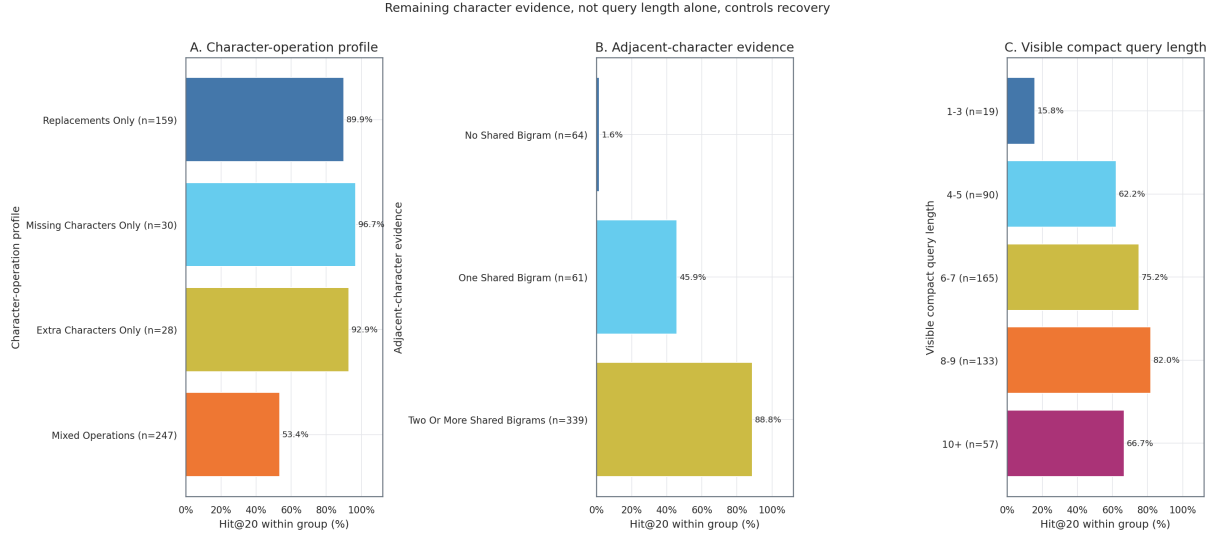


Figure 5: All panels report Hit@20 within the printed group denominator. Panel A groups source alignment operations, panel B groups shared compact bigrams, and panel C groups visible compact query length.

Insights

1. Mixed insertion, deletion, and replacement rows reach 53.44% Hit@20 and create 115 misses. (They account for 85.82% of all primary misses.)
2. No shared bigram reaches 1.56% Hit@20, one bigram 45.90%, and two or more 88.79%. (Adjacent-character evidence separates recoverable and unrecoverable queries more sharply than length alone.)
3. One-to-three-character inputs have the worst rate, 15.79% Hit@20, but six-to-seven-character inputs create more misses, 41, because that group is larger. (Failure rate and failure count answer different questions.)

OCR input	Verified family	A4 first family	Rank	Failure interpretation
MYMLAX	MYOLAX	MYOLAX	1	One replacement; complete recovery.
KELOLAC	KETOROLAC	KETOLAC	2	Candidate exists; ranking error.
RIV	RIVOTRIL	RIVO	10	Visible fragment retains weak evidence.
GHONT	CALAMINE	GEOCAND	> 20	Seven edits remove target evidence.
H	RIVO	None	> 20	One character cannot identify a family.
RIVOTRIL	RIVO	RIVOTRIL	> 20	Exact different real-name collision.

Table 5: Worst rows are not one homogeneous error class. Ranking, retrieval, insufficient evidence, and label ambiguity require different product behavior.

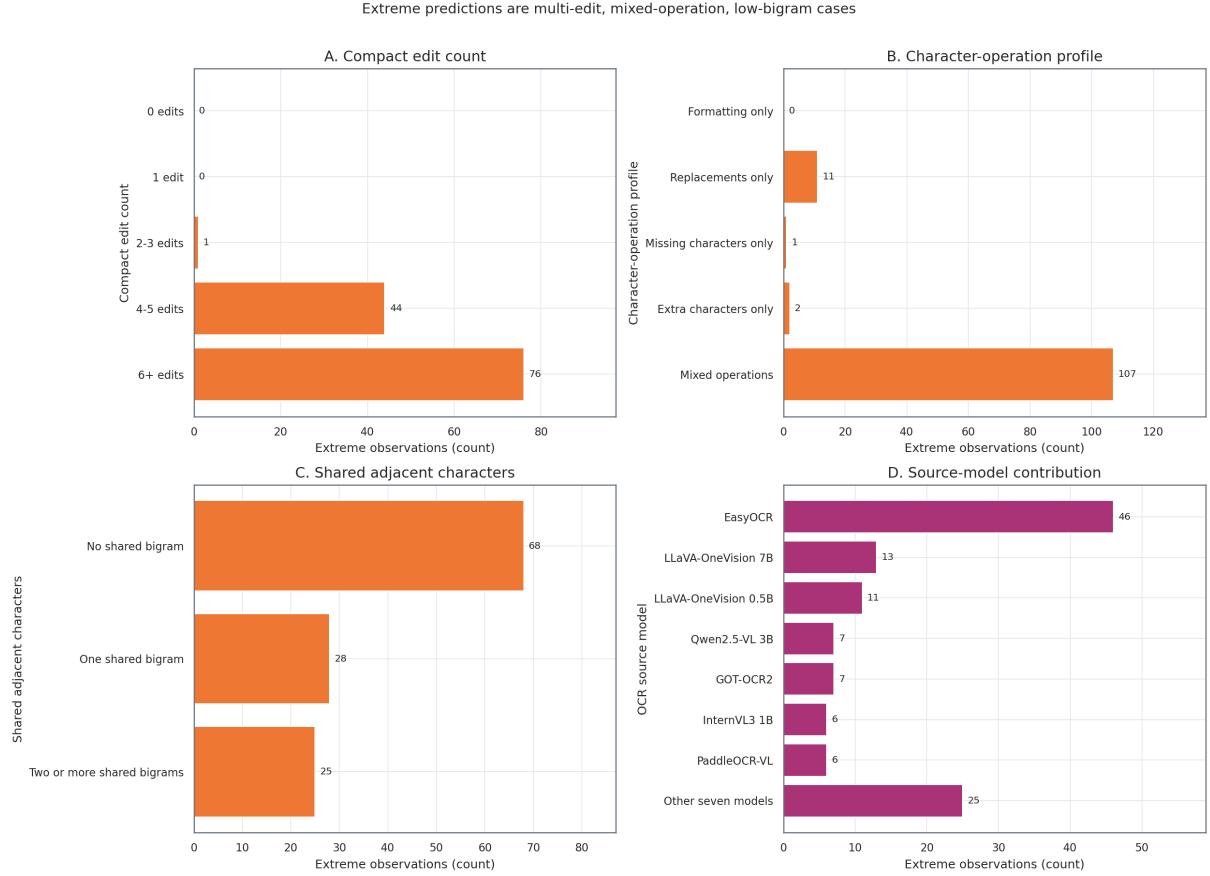


Figure 6: Each panel counts the 121 extreme observations, defined as compact distance divided by target length above 0.60 after collision and exact-match priority. Panels cover edit count, operation profile, shared bigrams, and source model.

Insights

1. The extreme cohort contains 76 six-plus-edit rows and 44 four-to-five edit rows. (It is a severe tail, not an extension of one-edit typos.)
2. Mixed operations account for 107 of 121 extreme rows, and 96 have at most one shared bigram. (Most rows lose both simple edit and local sequence evidence.)
3. Zero extreme rows reach rank 1, 10 reach ranks 2–20, and 111 miss the top 20. (Ordinary lexical retrieval rarely survives this evidence level.)

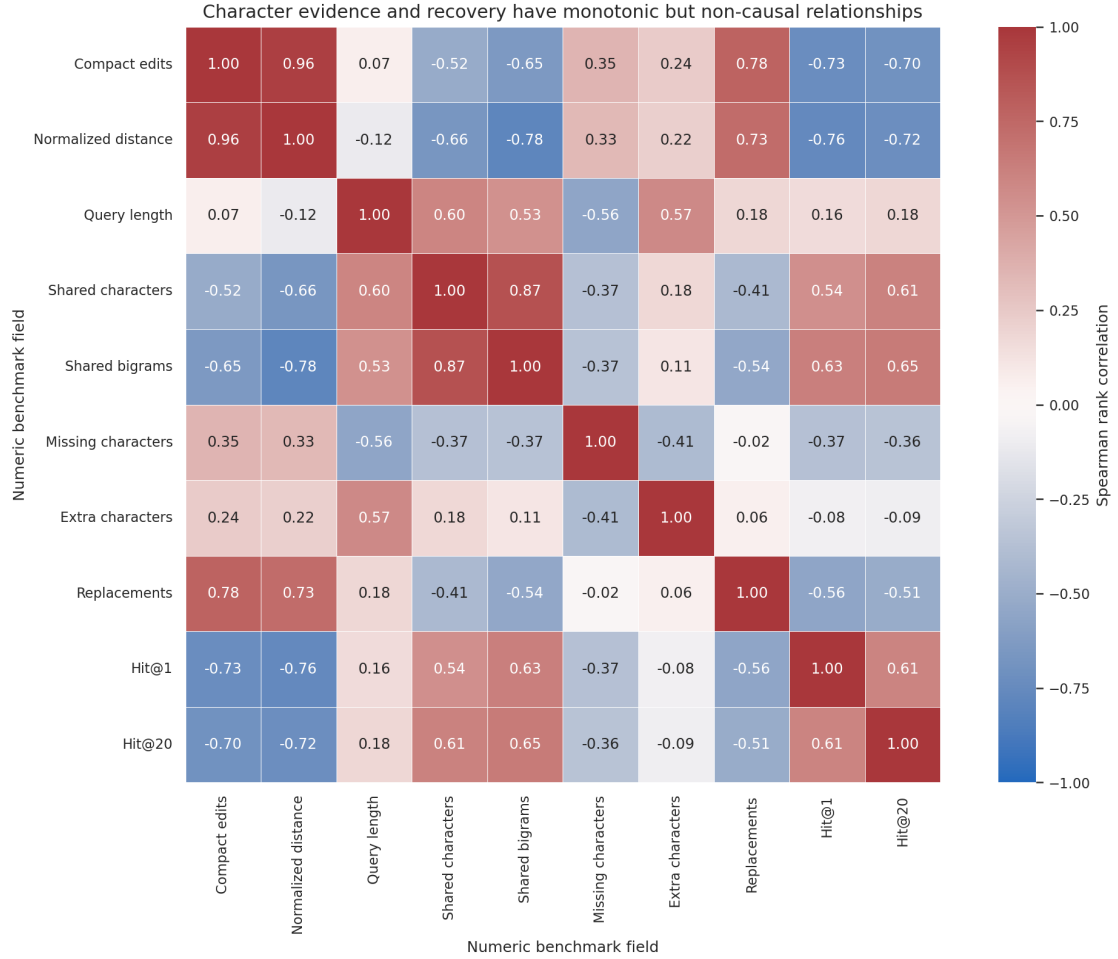


Figure 7: Cells are Spearman rank correlations over 464 primary pairs. Positive values mean two numeric fields tend to rise together; negative values mean one tends to fall as the other rises. Correlation does not prove causation.

Insights

1. Normalized distance has the strongest negative association with Hit@1, $\rho = -0.7592$, and Hit@20, $\rho = -0.7202$. (Larger corruption tracks lower recovery across the observed pairs.)
2. Shared bigrams correlate positively with Hit@1, $\rho = 0.6279$, and Hit@20, $\rho = 0.6520$. (Local sequence evidence is a useful candidate expansion and confidence feature.)
3. Replacement count correlates with Hit@20 at $\rho = -0.5100$. (The association includes overlap with total distance and mixed errors, so it is not an isolated causal effect.)

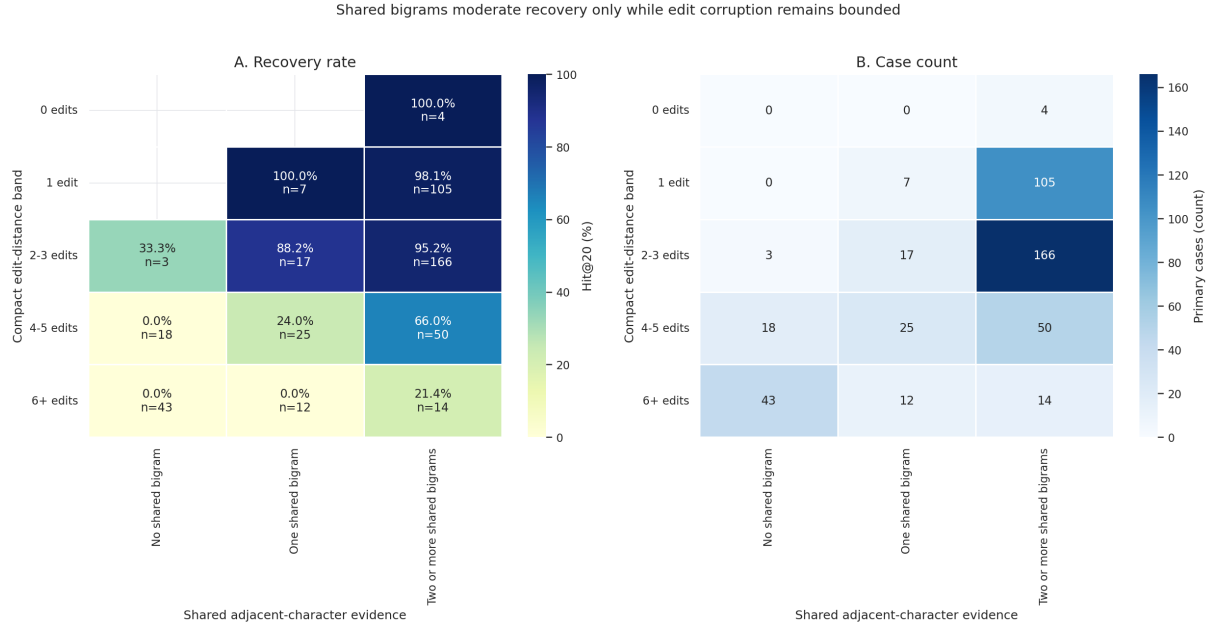


Figure 8: Rows are compact edit-distance bands and columns are shared-bigram bands. The left heatmap reports Hit@20 and case count in each cell; the right heatmap reports the corresponding denominator.

Insights

1. At four-to-five edits, Hit@20 is 0% with no shared bigram, 24% with one, and 66% with two or more. (Bigram evidence materially separates cases inside the same edit band.)
2. At six or more edits, even two-plus bigrams reach only 21.4% Hit@20. (Bigrams moderate bounded corruption but cannot replace missing lexical evidence in the severe tail.)

6 Retrieval-System Comparison

Ten systems receive the same 464 primary pairs, so differences are paired by query rather than compared across unrelated case sets.

System	Hit@1	Hit@5	Hit@20	MRR@20	Median ms
Algorithm 4, family rescue	47.6293%	61.8534%	71.1207%	0.537476	14.0704
Exhaustive Levenshtein	34.2672%	50.8621%	59.2672%	0.416955	0.7039
Jaro-Winkler	32.1121%	55.1724%	67.4569%	0.425369	0.6713
RapidFuzz token ratio	31.4655%	48.9224%	59.0517%	0.394378	8.0152
Algorithm 2, external fast	25.8621%	43.7500%	51.2931%	0.329160	3.8593
Algorithm 3, rank fusion	25.6466%	40.9483%	53.0172%	0.326419	6.1017
Algorithm 1, current app	23.7069%	31.4655%	38.1466%	0.272602	0.7372
Character 3-gram TF-IDF	18.3190%	35.7759%	47.4138%	0.257995	0.5042
Phonetic baseline	6.8966%	16.8103%	23.2759%	0.110216	0.4722
Exact or prefix match	2.5862%	2.8017%	3.0172%	0.027155	0.7060

Table 6: Best retrieval values are bold. Query latency excludes preparation; Algorithm 4 prepares its combined index in 8,731.705 ms before warm queries.

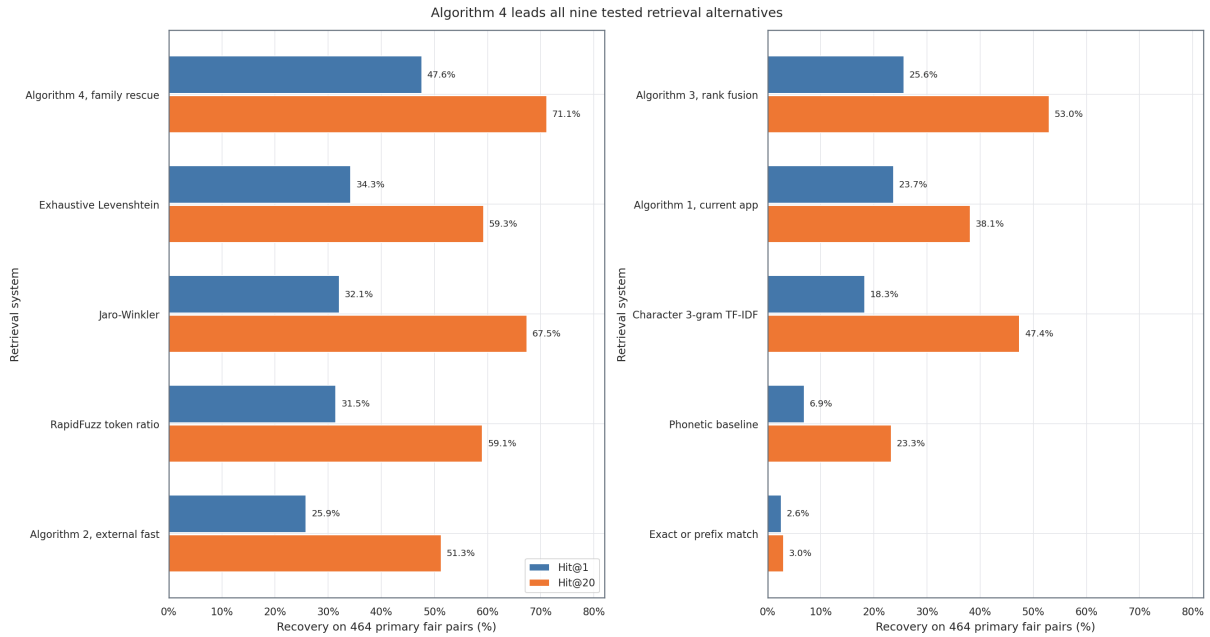


Figure 9: Blue bars are Hit@1 and orange bars Hit@20 over the same 464 primary pairs. Systems are split into two panels only to preserve readable labels; the x-axis scale is shared.

Insights

1. Algorithm 4 leads all nine alternatives at both cutoffs. (Its 47.6293% Hit@1 is 13.3621 points above exhaustive Levenshtein.)
2. Jaro-Winkler is the strongest classical Hit@20 baseline at 67.4569%. (Algorithm 4 adds 3.6638 points, so its main advantage over Jaro-Winkler is not unlimited candidate recall.)
3. Exact/prefix and phonetic-only systems fail on this OCR error mix. (Their Hit@20 values are 3.0172% and 23.2759%.)

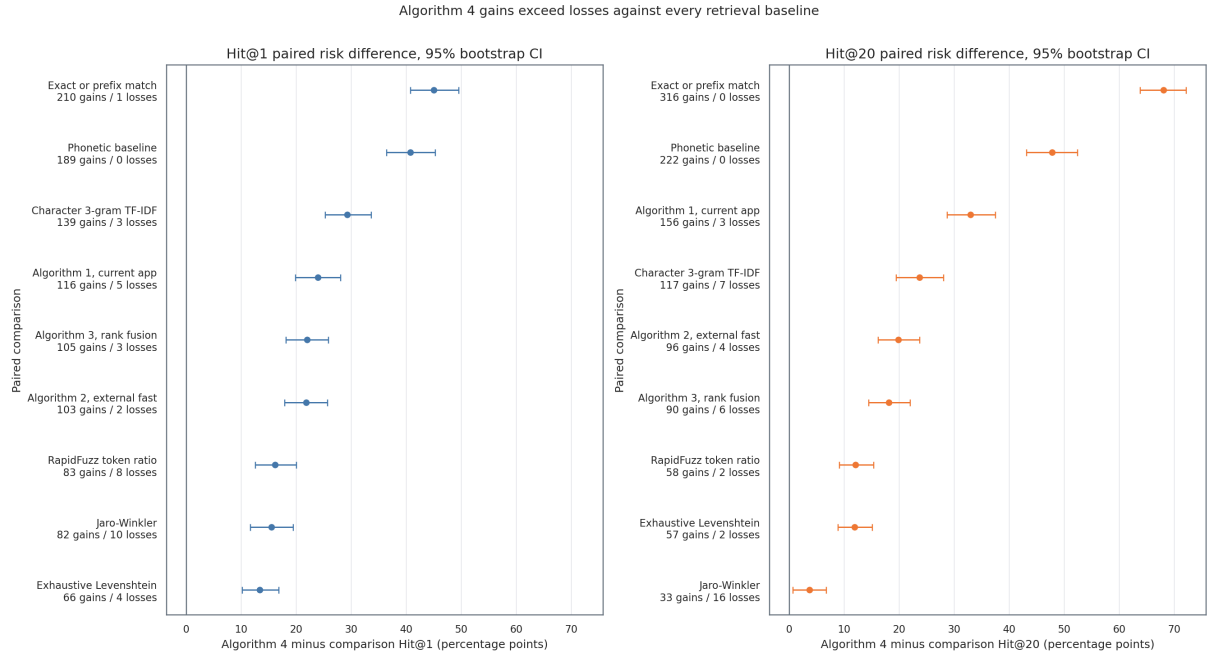


Figure 10: Each point is Algorithm 4 minus one comparison on the same 464 pairs; horizontal bars are 95% paired bootstrap intervals. Labels show cases won by Algorithm 4 and cases won by the comparison.

Insights

1. Against exhaustive Levenshtein, Algorithm 4 gains 66 and loses four Hit@1 pairs. (Difference +13.3621 points, 95% CI [10.1293, 16.8103], exact McNemar $p = 1.65 \times 10^{-15}$.)
2. Against Jaro-Winkler at Hit@20, Algorithm 4 gains 33 and loses 16. (Difference +3.6638 points, 95% CI [0.6466, 6.6810], $p = 0.0213$, matched odds ratio 2.03.)
3. Against Algorithm 3, Algorithm 4 gains 105 and loses three Hit@1 pairs. (The OCR ranking architectures are not equivalent even though both use multiple retrieval signals.)

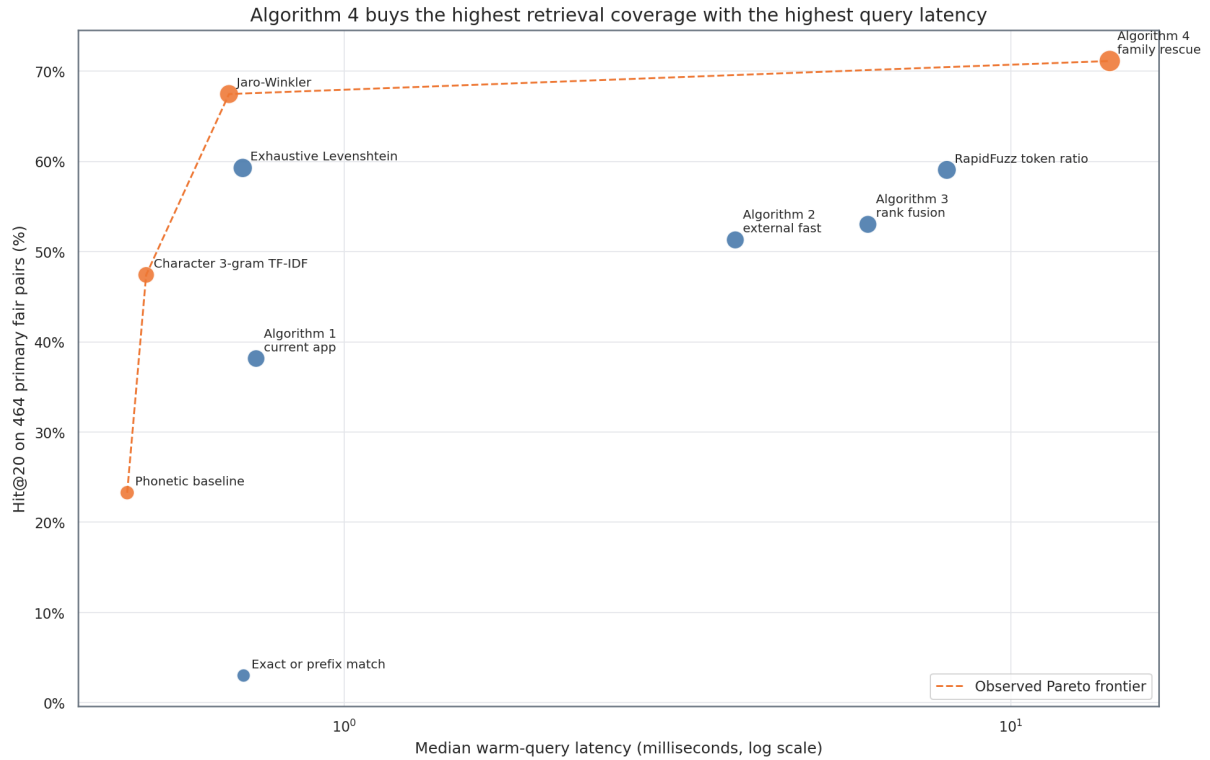


Figure 11: The x-axis is median warm-query latency on a logarithmic scale; the y-axis is Hit@20. Point size encodes Hit@1, and orange points form the observed latency-recovery Pareto frontier.

Insights

1. Algorithm 4 has the highest Hit@20 and the highest median query latency, 14.0704 ms. (It trades roughly 20 times Levenshtein latency for 11.8535 additional Hit@20 points.)
2. Jaro-Winkler, character 3-gram TF-IDF, the phonetic baseline, and Algorithm 4 lie on the observed frontier. (Each occupies a distinct speed-coverage point among the tested implementations.)
3. The frontier is machine- and implementation-specific. (It measures this single-process benchmark, not production throughput under concurrency.)

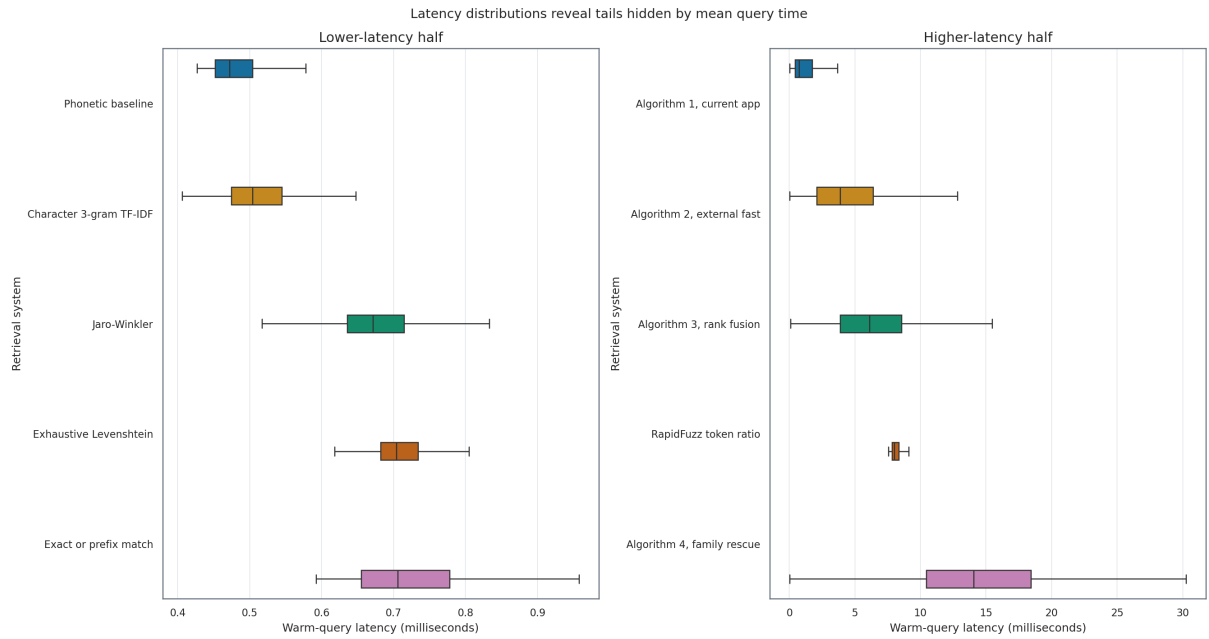


Figure 12: Each box plot shows warm-query latency over 464 primary pairs; the center line is the median, the box spans the middle 50%, and whiskers exclude displayed outliers. Systems are split into lower- and higher-latency panels.

Insights

1. Median and mean are insufficient for deployment sizing because query latency has visible tails. (Algorithm 4 mean is 14.8852 ms and median 14.0704 ms; its measured 95th percentile is 30.1791 ms.)
2. RapidFuzz’s median is 8.0152 ms despite weaker retrieval than sub-1-ms Levenshtein and Jaro-Winkler. (Library name alone does not guarantee a faster implementation under this workload.)

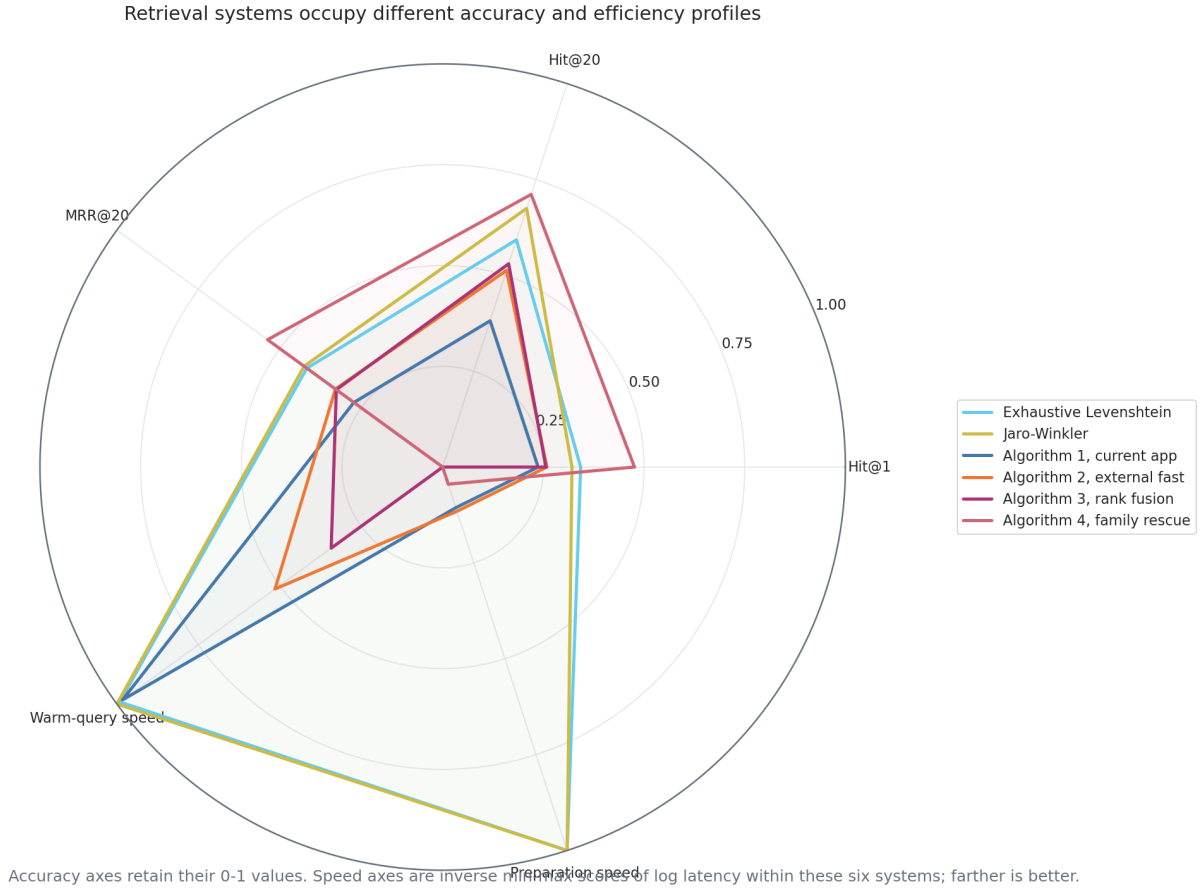


Figure 13: Six representative systems are compared on Hit@1, Hit@20, MRR@20, warm-query speed, and preparation speed. Accuracy axes keep their 0–1 values; speed axes are inverse min-max scores of log latency within these six systems, so farther from the center is better but relative.

Insights

1. Algorithm 4 dominates the three recovery axes but not either efficiency axis. (The chart makes the accuracy-cost trade-off explicit rather than compressing it into one score.)
2. The radial speed values are not percentages and cannot be compared outside this six-system set. (They are visualization-only min-max transforms; Table 6 contains actual milliseconds.)

Algorithm 4 recovers many cases missed by strong alternatives, but not all

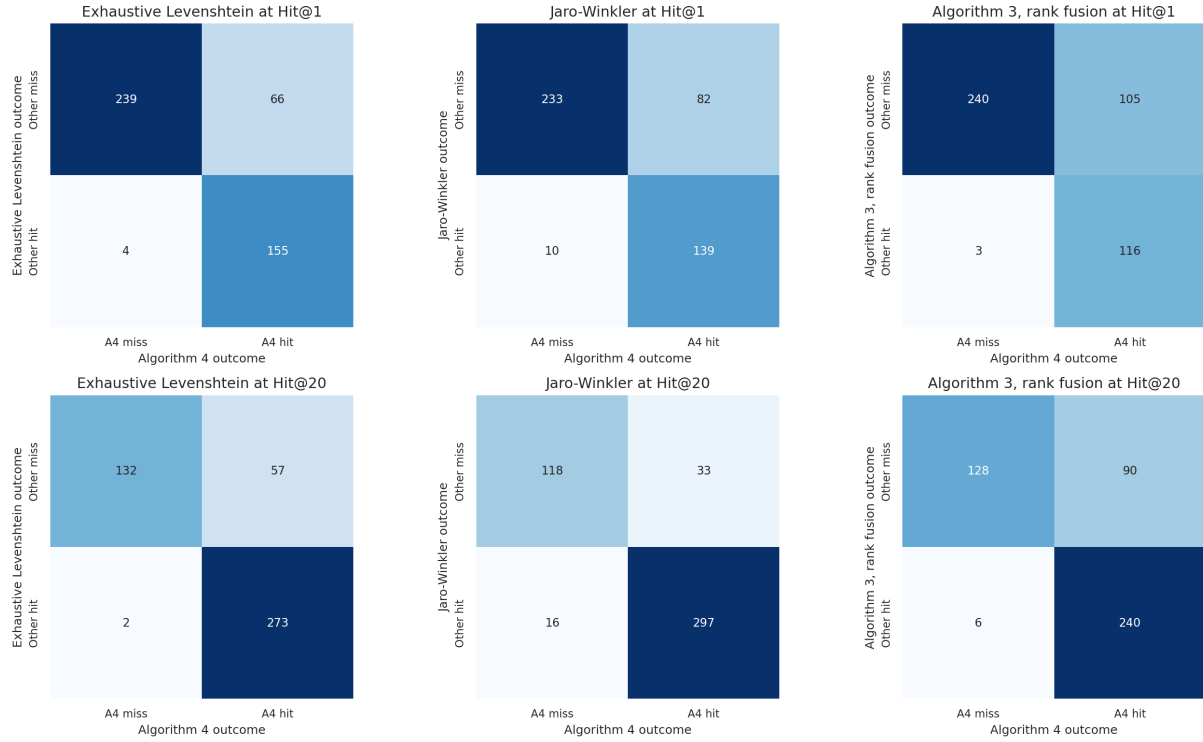


Figure 14: Each matrix cross-tabulates another system’s hit or miss against Algorithm 4 on the same pair. Columns are Algorithm 4 outcomes, rows are the comparison outcomes, and cells are pair counts.

Insights

1. Algorithm 4 uniquely recovers 66 Hit@1 pairs missed by Levenshtein, while Levenshtein uniquely recovers four. (Most of the aggregate gain comes from real complementary cases, not reshuffling shared successes.)
2. At Hit@20, Jaro-Winkler uniquely recovers 16 pairs that Algorithm 4 misses. (The strongest classical retriever still contains useful recall not captured by the combined system.)
3. Algorithm 4 uniquely recovers 90 Hit@20 pairs over Algorithm 3 and loses six. (Family rescue changes candidate presence, not only top-rank score.)

7 Algorithm 4 Enhancement

The enhancement converts observed failure patterns into bounded, general rules without storing query-specific answers or benchmark medicine names.

Rule	General behavior	Concrete recovery
Three-character delete-key retrieval	Permit bounded deletion-index candidates for compact queries of length three.	RIO $\rightarrow$ RIVO.
Short-edge retrieval	Search conservative prefix and suffix indexes with fixed core and rescue caps.	Retains evidence when only an edge is readable.
Family-head rescue	Compare a corrupted query with the shared head of commercial variants.	MYOLANA $\rightarrow$ MYOLAX.
Compatible-length scan	Expand only to catalog families in plausible query-length buckets.	LGCMU $\rightarrow$ LACTO.
Strict full-name correction	Allow a rescue result to replace first only for length at least five, raw distance at most three, strictly better raw distance, no worse weighted distance, and score gap at most 0.35.	NIGROHEX $\rightarrow$ VIGOREX; NEITROSET $\rightarrow$ NEUROCET.
Clarification state	Keep equal evidence, family variants, real-name collisions, and low-evidence strings non-confident.	CONAL compares COBAL and CONIL.

Table 7: The rules depend on query shape and catalog evidence. No literal test query or expected medicine was inserted into Algorithm 4.

Locked suite	Cases	H@1 before	H@1 after	H@20 before/after	Paired guardrail
Manual failed cases	50	72.00%	74.00%	92.00% / 94.00%	No unsafe confident result.
Primary OCR pairs	464	46.34%	47.63%	70.91% / 71.12%	H@1 gains/losses 6/0; H@20 1/0.
Browser primary pairs	464	37.93%	40.09%	62.07% / 63.58%	H@1 gains/losses 10/0; H@20 7/0.
Proportional synthetic sample	6,000	81.12%	81.40%	92.83% / 93.28%	Behavior after 93.45%.
Full synthetic benchmark	115,000	81.86%	82.12%	92.99% / 93.41%	H@1 gains/losses 180/95; H@20 132/5.

Table 8: The final rules were retained only after broader suites showed net gains without unsafe confident first-result regression. Browser values remain separate because JavaScript and Python retrieval implementations differ.

8 Algorithm 4 Ablation

Each ablation removes one named query-time component while holding the 464 primary pairs, catalog, and scoring contract fixed.

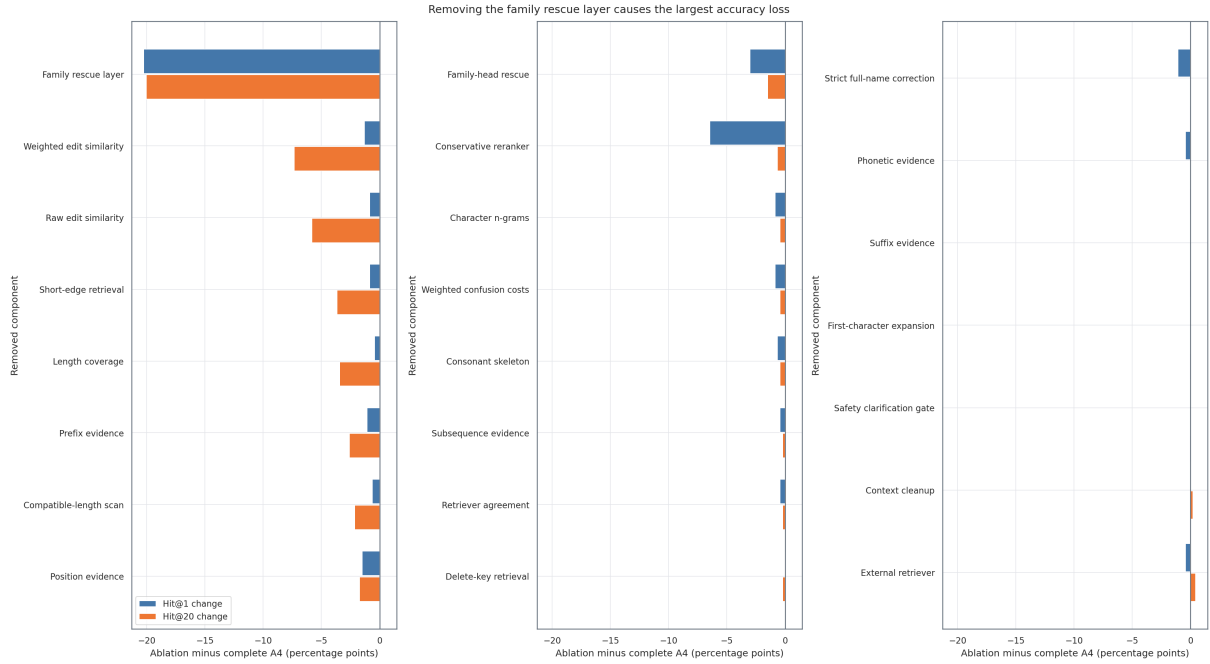


Figure 15: Each row removes one Algorithm 4 component. Blue is Hit@1 change and orange Hit@20 change relative to complete Algorithm 4; negative values mean the complete system performs better.

Insights

1. Removing family rescue produces the largest loss, -20.2586 Hit@1 points and -20.0431 Hit@20 points. (It is the main source of candidate recovery beyond the child algorithms.)
2. Weighted edit similarity causes the next-largest Hit@20 loss, -7.3276 points; raw edit similarity loses -5.8190 points. (Both distance views contribute non-redundant retrieval evidence.)
3. Removing the external retriever or context cleanup raises Hit@20 by only 0.4310 and 0.2155 points. (Paired tests are not significant, so these small reversals require larger confirmation before deletion.)

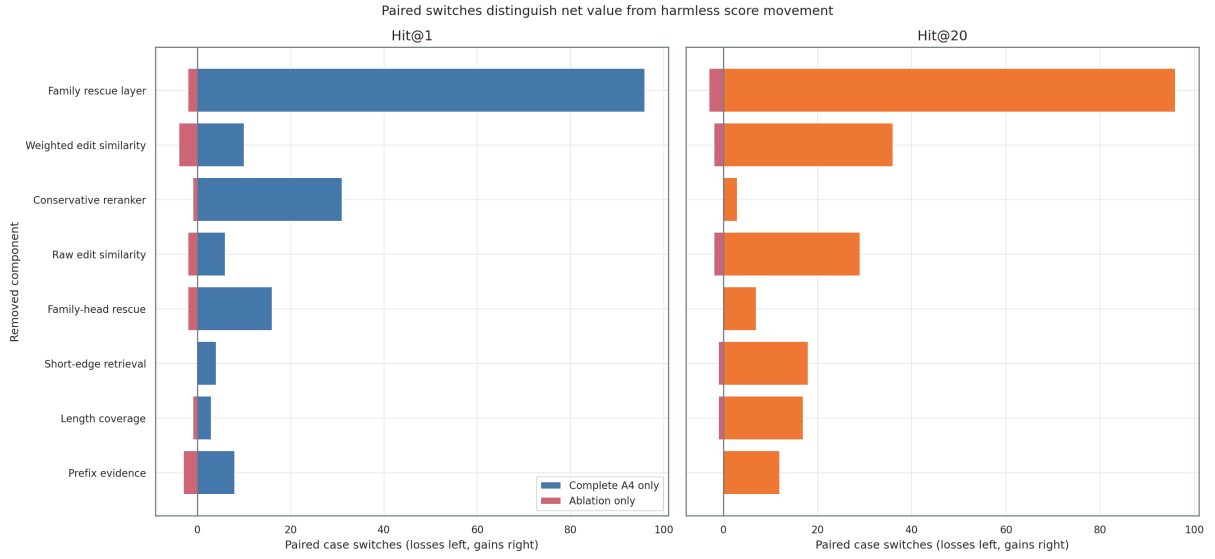


Figure 16: For each removed component, blue counts primary pairs where complete Algorithm 4 succeeds and the ablation fails; orange counts the reverse. Panels separate Hit@1 and Hit@20 switches.

Insights

1. Complete Algorithm 4 beats no-rescue on 96 pairs and loses on two at Hit@1; at Hit@20 the counts are 96 and three. (Exact McNemar p -values are below 10^{-24} .)
2. The conservative reranker changes first rank far more than candidate recall: 31 gains versus one loss at Hit@1, but only three gains at Hit@20. (Its role is ordering, not broad retrieval.)
3. Strict full-name correction yields five Hit@1 gains and zero losses, with exact $p = 0.0625$. (Direction is consistent across development and holdout, but this sample is too small for a 0.05 claim.)

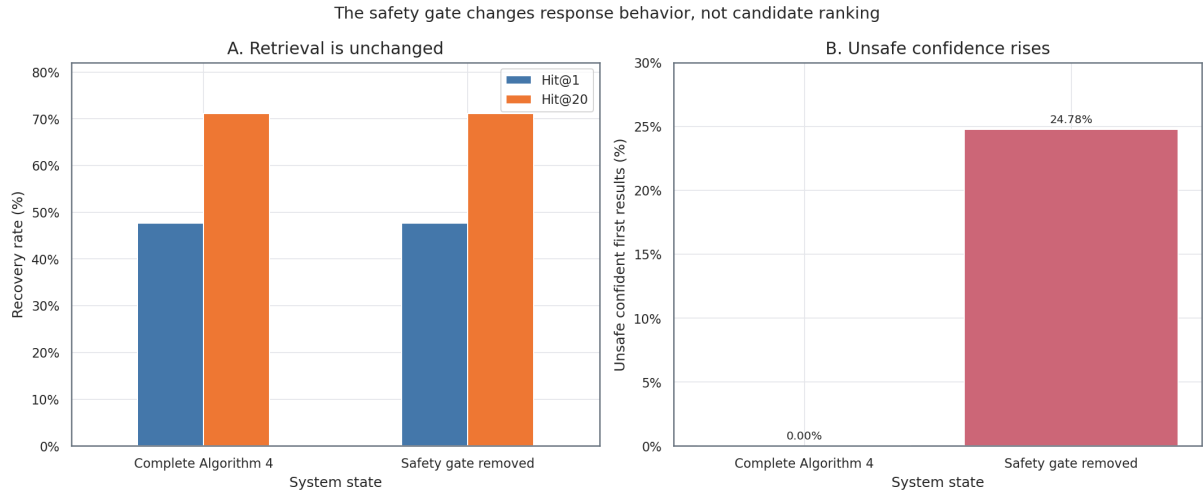


Figure 17: The left panel compares Hit@1 and Hit@20 with and without the safety clarification gate; the right panel compares unsafe confident first-result rate. Both variants use identical ranked candidates.

Insights

1. Retrieval metrics remain exactly 47.6293% Hit@1 and 71.1207% Hit@20 after gate removal. (The gate operates on response confidence, not rank.)
2. Unsafe confident first results rise from 0% to 24.7845%. (A retrieval ablation alone would falsely label the safety gate as unnecessary.)

Table 9: Complete 24-configuration ablation result on 464 primary pairs. Deltas are percentage points relative to complete Algorithm 4.

Configuration	Hit@1	Hit@20	Δ H@1	Δ H@20
Complete Algorithm 4	47.63%	71.12%	0.00	0.00
Without external retriever	47.20%	71.55%	-0.43	+0.43
Without context cleanup	47.63%	71.34%	0.00	+0.22
Without family rescue	27.37%	51.08%	-20.26	-20.04
Without raw edit similarity	46.77%	65.30%	-0.86	-5.82
Without weighted edit similarity	46.34%	63.79%	-1.29	-7.33
Without prefix evidence	46.55%	68.53%	-1.08	-2.59
Without suffix evidence	47.63%	71.12%	0.00	0.00
Without character n-grams	46.77%	70.69%	-0.86	-0.43
Without phonetic evidence	47.20%	71.12%	-0.43	0.00
Without consonant skeleton	46.98%	70.69%	-0.65	-0.43
Without subsequence evidence	47.20%	70.91%	-0.43	-0.22
Without position evidence	46.12%	69.40%	-1.51	-1.72
Without length coverage	47.20%	67.67%	-0.43	-3.45
Without delete-key retrieval	47.63%	70.91%	0.00	-0.22
Without short-edge retrieval	46.77%	67.46%	-0.86	-3.66
Without first-character expansion	47.63%	71.12%	0.00	0.00
Without compatible-length scan	46.98%	68.97%	-0.65	-2.16
Without family-head rescue	44.61%	69.61%	-3.02	-1.51
Without weighted confusion costs	46.77%	70.69%	-0.86	-0.43
Without retriever agreement	47.20%	70.91%	-0.43	-0.22
Without strict full-name correction	46.55%	71.12%	-1.08	0.00
Without conservative reranker	41.16%	70.47%	-6.47	-0.65
Without safety clarification gate	47.63%	71.12%	0.00	0.00

9 Equal Edit-Distance Ordering

Equal raw edit distance leaves 37 primary ranking failures unresolved, so eight generic evidence policies were tested without privileging a specific character position.

Input	Verified	Current first	Evidence and decision
LAMIX	LASIX	VAMIX	Both raw distances are 1; expected rank is 3. Position and edge evidence disagree, so both candidates remain visible.
CONAL	COBAL	CONIL	Both raw distances are 1; expected rank is 3. The first character matches every candidate and cannot resolve the tie.
OSMOL	OSTOCAL	OSMO PEEL	Both raw distances are 3; expected rank is 16. Equal edit count hides different length and family evidence.

Table 10: The product response for these cases is candidate comparison and clarification, not a hidden alphabetical or first-character rule.

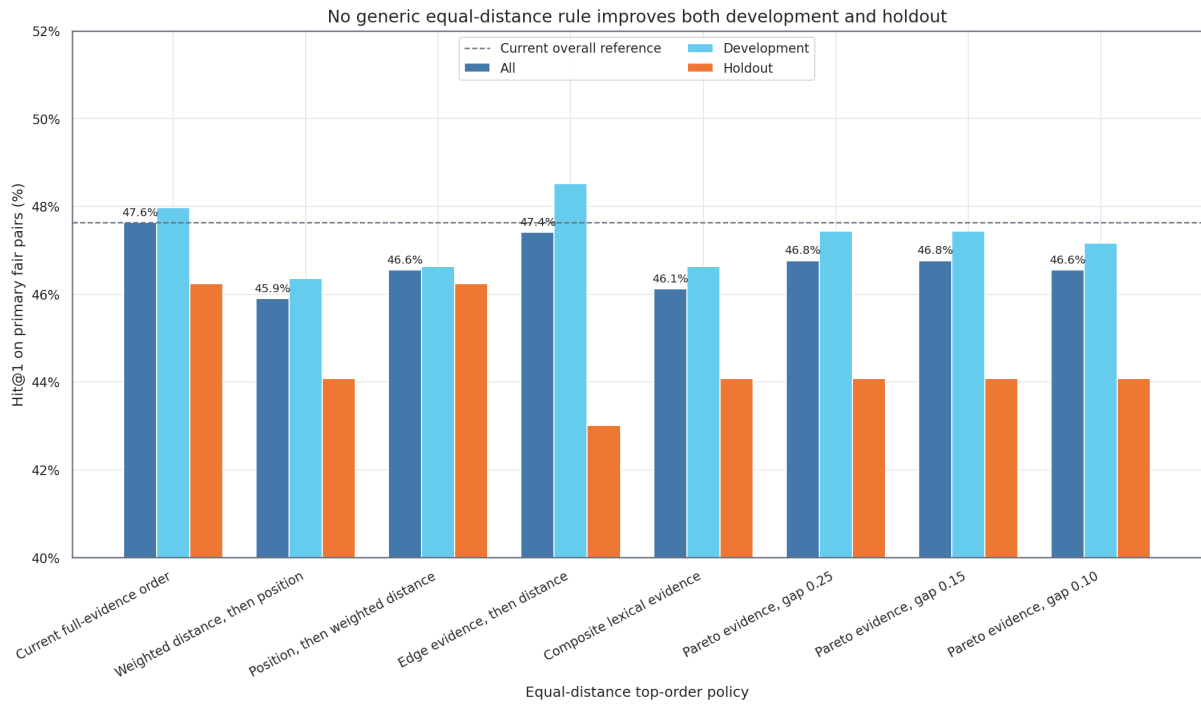


Figure 18: The y-axis is Hit@1; each policy has all, development, and holdout bars. The dashed line is current overall ordering. Target-family hashing gives 371 development pairs and 93 disjoint holdout pairs.

Insights

1. No replacement policy improves both development and holdout. (Current ordering is 47.63% overall, 47.98% development, and 46.24% holdout.)
2. Edge-first ordering rises to 48.52% on development but falls to 43.01% on holdout. (Choosing it from development would overfit the selected target families.)
3. Position-first ties current holdout but reduces development and overall accuracy. (Matching the first character is useful evidence, not a safe universal decision chain.)

10 Synthetic 115,000-Case Benchmark

The deterministic synthetic benchmark measures four algorithms over 34 named mutation categories, four scopes, safety behavior, and failure mechanism.

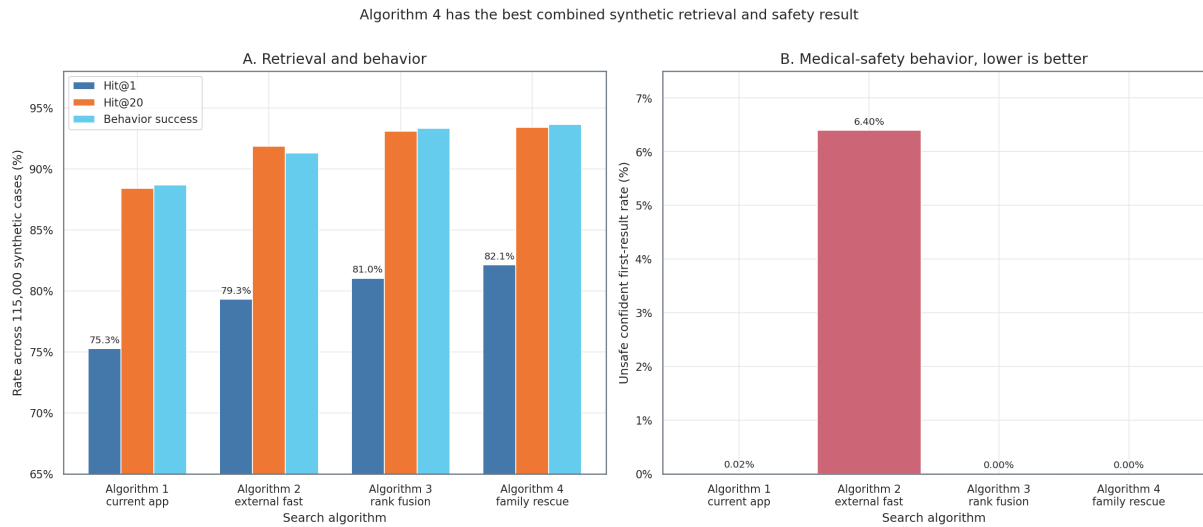


Figure 19: Panel A compares Hit@1, Hit@20, and behavior success over all 115,000 cases; panel B reports unsafe confident first-result rate, where lower is better. Algorithms 1–4 use aligned source rows.

Insights

1. Algorithm 4 reaches 82.1165% Hit@1, 93.4122% Hit@20, and 93.6374% behavior success. (All are the best aggregate values among the four.)
2. Algorithm 2 reaches 79.3009% Hit@1 but produces 6.3957% unsafe confident first results. (High retrieval without clarification control is unacceptable for the intended interface.)
3. Algorithm 4 versus Algorithm 3 gains 6,701 and loses 5,457 Hit@1 rows. (Net gain 1,244 rows, +1.0817 points, 95% CI [0.8913, 1.2739].)

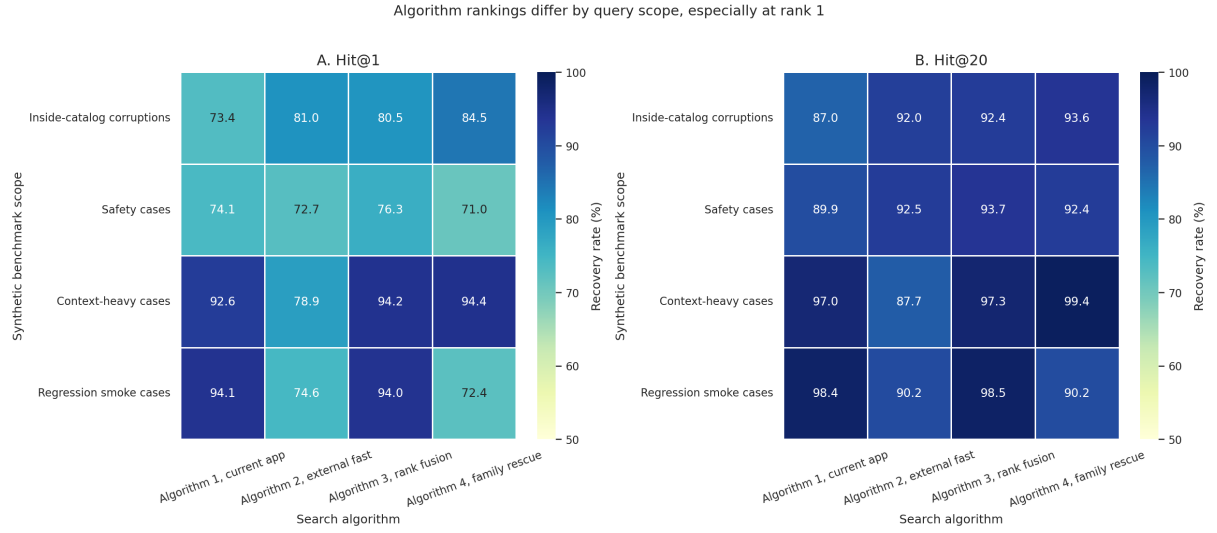


Figure 20: Rows are inside-catalog corruptions, safety cases, context-heavy queries, and regression smoke tests; columns are Algorithms 1–4. The left heatmap is Hit@1 and the right is Hit@20.

Insights

1. Algorithm 4 is strongest on inside-catalog Hit@1 at 84.5% and context-heavy Hit@20 at 99.4%. (Family rescue and context cleanup match those query shapes.)
2. Algorithm 4 does not lead safety Hit@20 or smoke Hit@20. (Algorithm 3 reaches 93.7% and 98.5% respectively, while Algorithm 4 reaches 92.4% and 90.2%.)
3. Aggregate leadership therefore hides regressions in specific scopes. (Release criteria must keep scope-level floors, not only one overall target.)

Type	Input → expected	A4 result	Measured state	Correct treatment
1	epigent → APIGENT	Exact EPIGENT first	Real-name collision	Exclude from fair single-answer accuracy; keep safety diagnostic.
2	conal → COBAL	CONIL first; expected rank 3	Equal edit evidence	Compare candidates; do not force one position rule.
3	acc plus unreadable-after → ACCELAFEN	Expected rank 4	Known missing continuation	Use user/image location evidence.
4	candeblockb → CANDEBLOCK D	Family head first	Family-variant mismatch	Recover family, then choose form or variant.
5	akc → ACC	No target in top 20	Candidate-generation failure	Expand bounded retrieval or ask for more evidence.
6	tidanhair → TITAN HAIR	Expected rank 2	Candidate-ranking failure	Improve rank evidence while retaining clarification.

Table 11: The six mechanisms are orthogonal to the 34 mutation categories and map directly to different algorithm or interface actions.

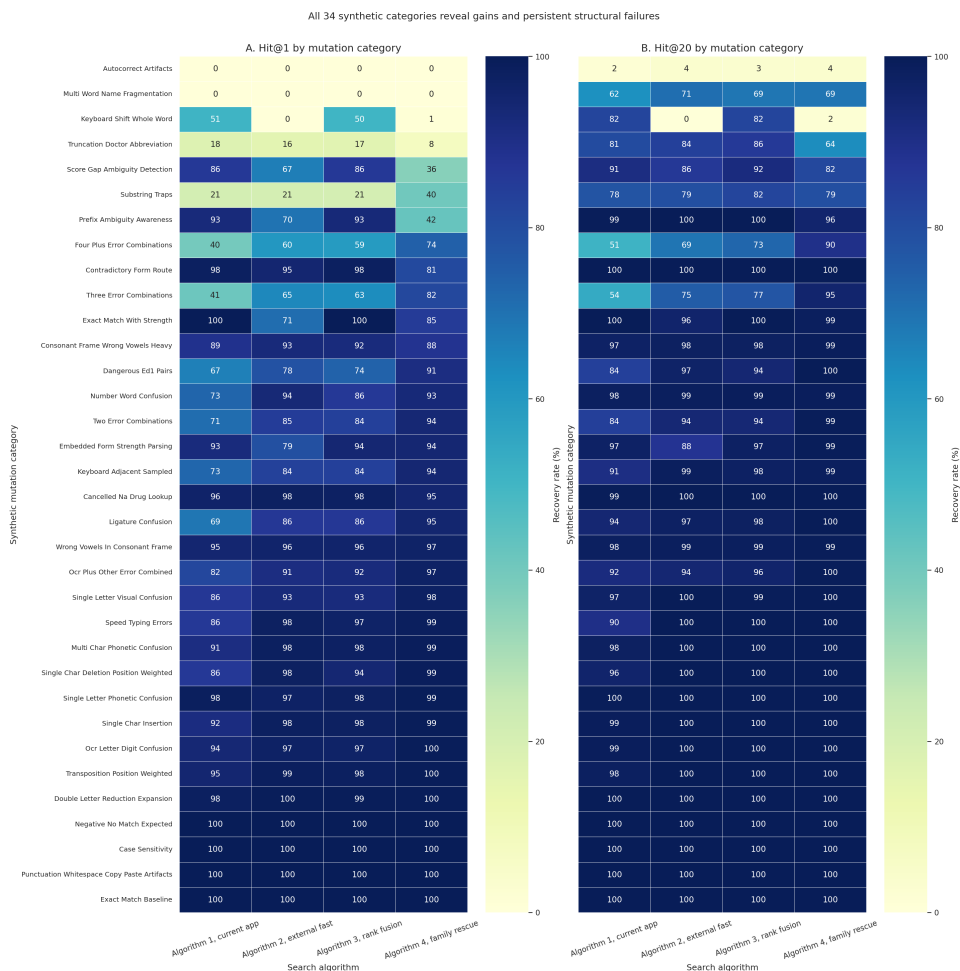


Figure 21: Each row is one of 34 mutation categories ordered by Algorithm 4 Hit@1; columns are Algorithms 1–4. The left heatmap reports Hit@1 and the right Hit@20, both from 0% to 100%.

Insights

1. Algorithm 4 reaches near-complete recovery on many one-edit, OCR, punctuation, and exact-name categories. (These rows supply character evidence compatible with its indexes and scoring signals.)
2. Autocorrect artifacts remain effectively unrecoverable, 0% Hit@1 and 3.65% Hit@20. (Inputs such as a different English word can contain no lexical evidence for the expected medicine.)
3. Keyboard shift and doctor abbreviation remain structural weaknesses. (Algorithm 4 Hit@20 is 2.0% and 63.5%; the first needs a dedicated reversible decoder, while the second needs cautious prefix retrieval.)
4. Multi-word fragmentation has 0% Hit@1 but 69.35% Hit@20. (Inputs such as PRO or IVY can be valid shorter catalog families, so forced first-rank scoring conflicts with real ambiguity.)

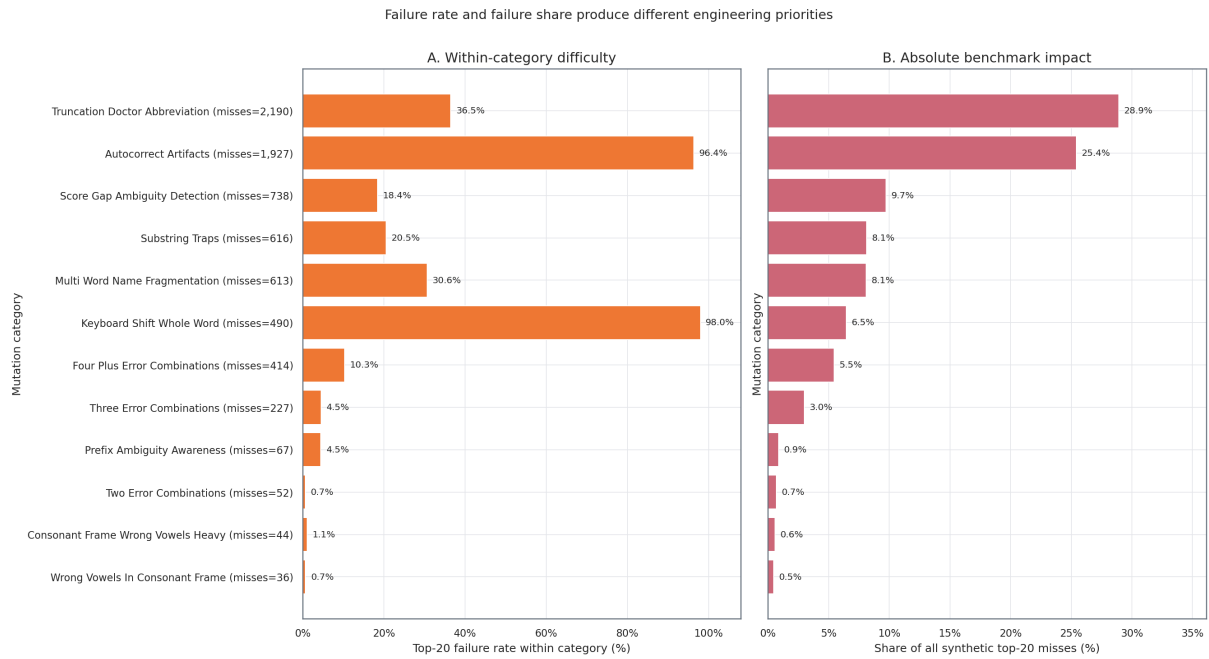


Figure 22: The left panel is top-20 failure rate inside each of the 12 largest failure-contributing categories; the right panel is each category’s share of all Algorithm 4 top-20 misses. Labels print category size or miss count.

Insights

1. Doctor abbreviations create 2,190 misses, 28.91% of all top-20 misses, while autocorrect artifacts create 1,927, 25.44%. (Together they cause more than half of the benchmark’s retrieval failures.)
2. Keyboard shift has a 98% failure rate but only 6.47% failure share because it contains 500 rows. (A high rate does not automatically make a category the largest engineering target.)
3. The six largest categories create 86.77% of all misses. (Prioritizing doctor abbreviation, autocorrect, score-gap ambiguity, substring traps, fragmentation, and keyboard shift covers most measured failure impact.)



Figure 23: Panel A counts diagnostic or failed rows assigned to each search mistake mechanism; panel B reports Hit@20 inside that mechanism. Mutation category describes how input was generated, while mistake type describes why search did not return the intended first answer.

Insights

1. Candidate-ranking failure is largest at 7,279 rows and has 100% Hit@20. (The expected family already exists in the list, so reranking and family presentation can address it.)
2. Candidate-generation failure contains 3,794 rows and 0% Hit@20. (No reranker can fix a target absent from the candidate set.)
3. Known unreadable continuation reaches 65.63% Hit@20. (User-supplied position evidence before, after, or in the middle can narrow candidates, but it cannot manufacture missing characters.)

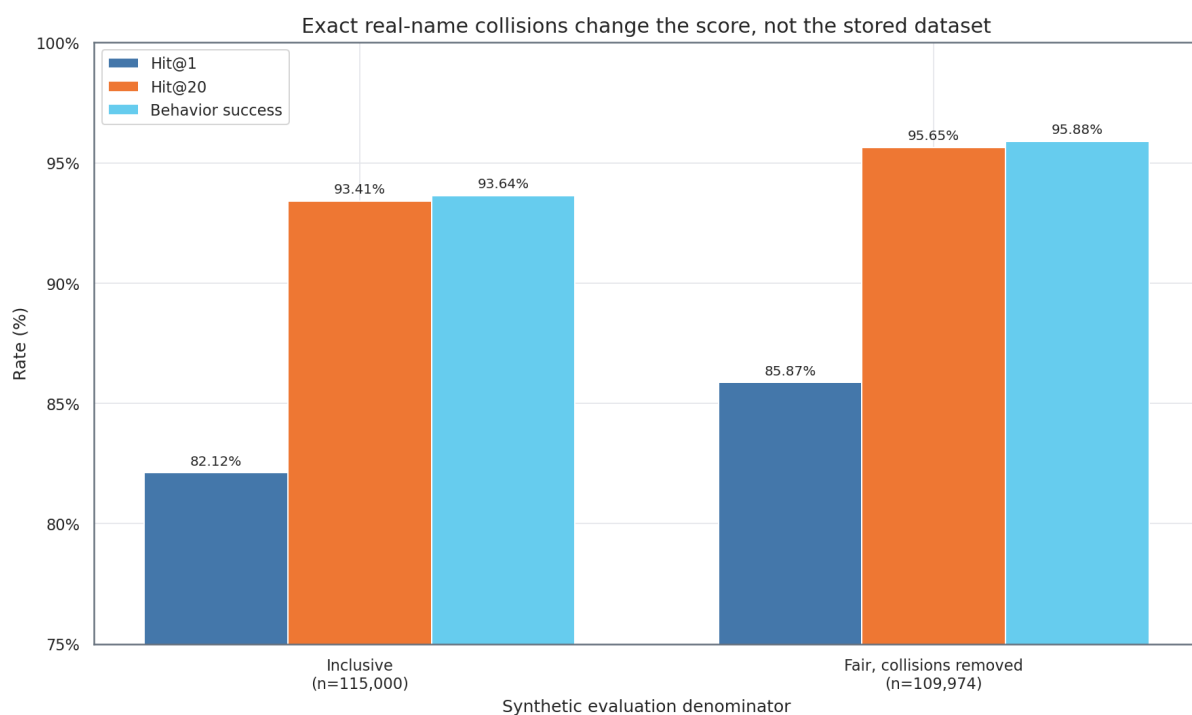


Figure 24: Each group reports Hit@1, Hit@20, and behavior success. Inclusive uses all 115,000 rows; fair removes 5,026 exact-real-name collision rows but retains them in the stored benchmark and diagnostic reports.

Insights

1. Fair Hit@1 is 85.8694%, 3.7529 points above the inclusive 82.1165%. (Exact different real names make a unique intended answer unavailable from text alone.)
2. Fair Hit@20 is 95.6490%, 2.2368 points above inclusive 93.4122%. (Both denominators remain necessary: fair for algorithm comparison, inclusive for supplied-data behavior.)

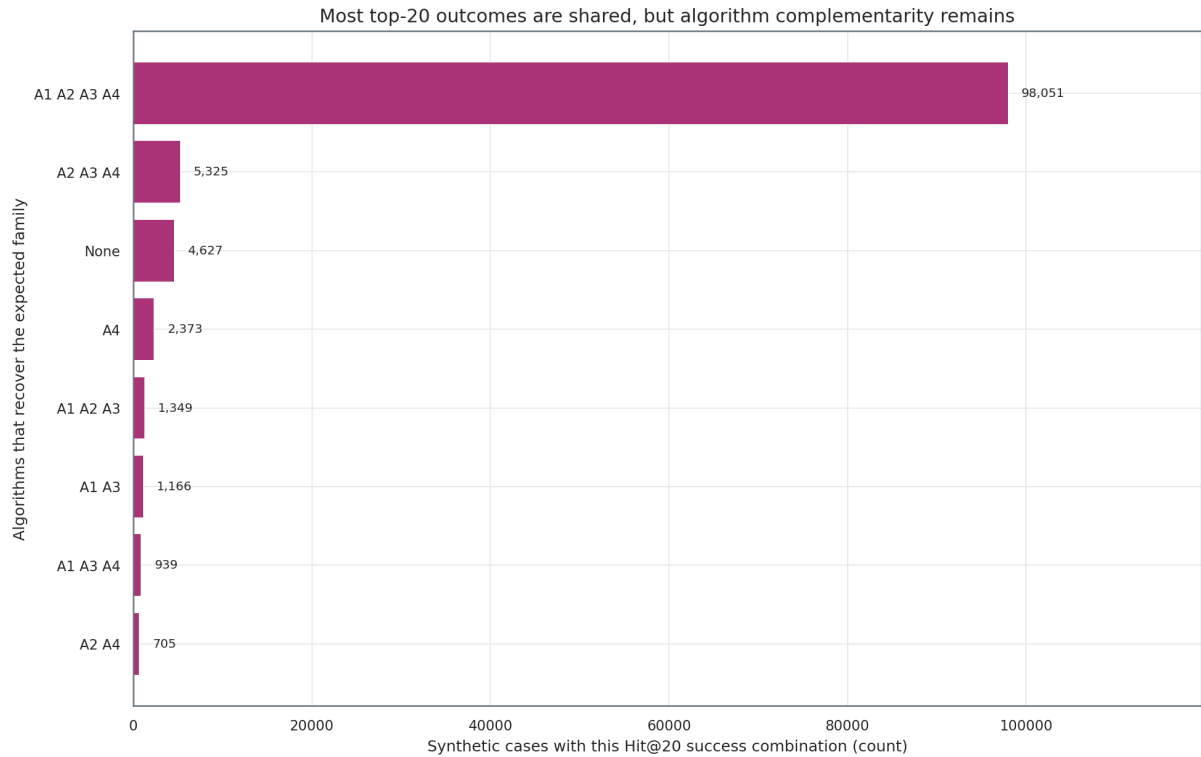


Figure 25: The y-axis names which Algorithms 1–4 recover the expected family by rank 20; the x-axis counts aligned synthetic rows with that exact success combination. The eight most frequent combinations are shown.

Insights

1. All four algorithms recover 98,051 cases, while none recovers 4,627. (Most outcomes are shared, and a hard residual remains outside every tested retrieval strategy.)
2. Algorithm 4 alone recovers 2,373 cases by rank 20. (Family rescue adds unique coverage beyond simple union behavior.)
3. Algorithms 2, 3, and 4 recover 5,325 cases missed by Algorithm 1. (The external retrieval path contributes broad complementary coverage.)

11 Pharmacist Study Status

The within-subject study is fully assigned but remains a protocol, not an outcome experiment.

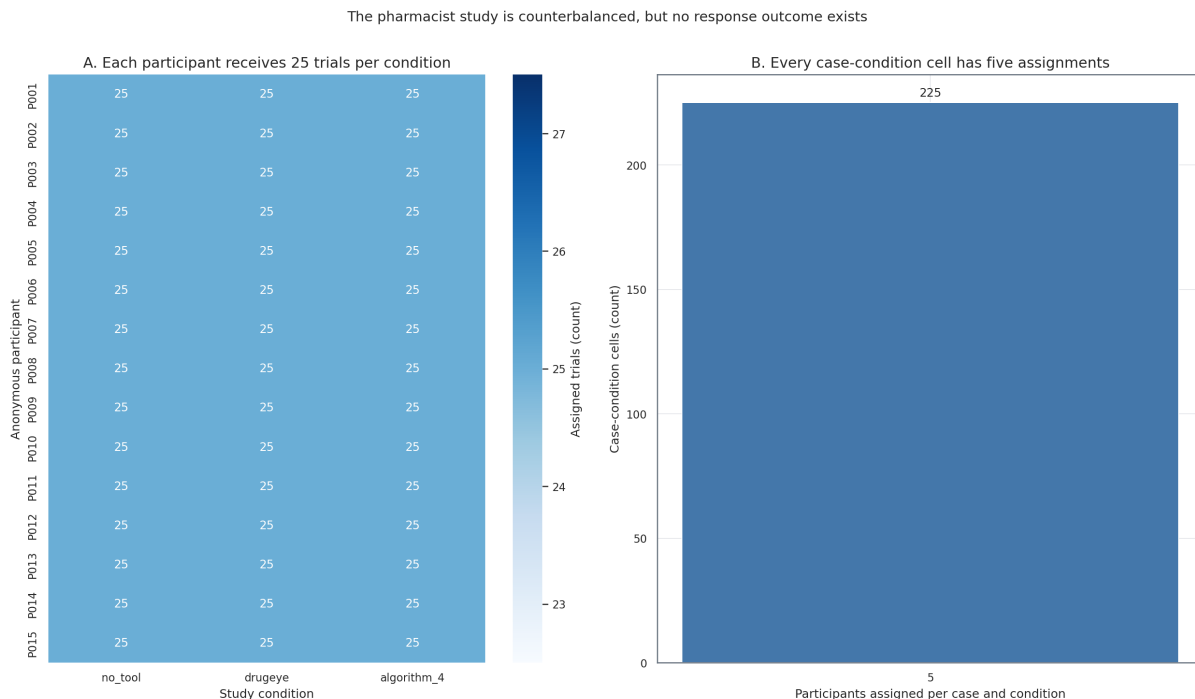


Figure 26: The left heatmap counts trials per participant and condition; the right counts presentations per case and condition. Conditions are no tool, DrugEye, and Algorithm 4. Colors encode assignment count, not performance.

Insights

1. Each of 15 participants receives 25 cases in each condition, 75 total. (The schedule contributes 375 decisions per condition and 1,125 overall.)
2. Each of 75 crops appears five times per condition and only once per participant. (The design balances condition exposure while avoiding a participant seeing the same crop twice.)
3. No accuracy, time, confidence, or unsafe-selection result exists. (Execution still requires de-identified crops, two-pharmacist target adjudication, ethics review, frozen DrugEye captures, and responses.)

12 Failures and Dead Ends

These findings identify work that failed, remains unsupported, or would produce misleading claims if reported as success.

- **Extreme lexical corruption remains unsolved.** Algorithm 4 misses 111 of 121 extreme observations by rank 20. (The cohort averages 6.16 compact edits and cannot be repaired reliably by small score changes.)
- **Autocorrect artifacts contain insufficient signal.** Algorithm 4 reaches 0% Hit@1 and 3.65% Hit@20. (Mappings such as a medicine to an unrelated English word require image evidence, a language model with constraints, or human clarification, not ordinary fuzzy matching.)
- **Keyboard-shift results are feature-specific.** Algorithm 4 reaches 0.8% Hit@1 and 2.0% Hit@20. (Algorithms 1 and 3 score much higher because they contain a whole-keyboard shift reversal; the category tests that decoder, not general spelling robustness.)

- **Generic equal-distance rules did not transfer.** Edge-first improved development by 0.54 points but reduced holdout by 3.23 points. (No policy replaced current ordering.)
- **Several ablation components remain unproven on 464 pairs.** Removing suffix evidence or first-character expansion causes no switches; removing the external retriever or context cleanup slightly raises Hit@20. (Zero or favorable removal effects are evidence for review, not proof that a component is universally useless.)
- **Raw OCR-model ranking is confounded.** Models did not process a matched image set. (Case severity, target distribution, and row count differ by model.)
- **DrugEye has no matched human-study outcome.** Historical web probes and cached sample results do not replace randomized participant decisions. (No clinical or workflow superiority claim is made.)

13 Open Questions and Next Steps

1. Audit repeated extreme OCR pairs against original crops, starting with exact alternative real names such as INDOMETHACIN and NEURONTIN; freeze corrected labels before further tuning.
2. Add bounded operation-aware candidate generation for four-to-five-edit rows with at least one shared bigram; evaluate on development, then require a holdout Hit@20 gain without increasing unsafe confidence.
3. Rerank ranks 2–5 using family-head, weighted edit, and agreement evidence; 89 of the 140 inclusive rank-only cases are resolved by rank 5.
4. Redesign autocorrect, fragmentation, doctor abbreviation, and keyboard shift categories so scoring reflects recoverable evidence and valid ambiguity rather than one forced expected family.
5. Run all 14 OCR systems on the same de-identified crop set with one preprocessing contract, then report exact OCR accuracy and downstream Algorithm 4 recovery separately.
6. Execute the pharmacist study only after two independent pharmacists adjudicate every crop; report correct selection, safe deferral, unsafe wrong selection, decision time, and confidence with participant- and case-clustered uncertainty.

14 Analysis Coverage and Reproducibility

The final audit distinguishes completed analyses from analyses the available files cannot support.

- The analysis script reads every designated source in full, validates 595 OCR observations, 464 primary pairs, 115,000 aligned synthetic rows per algorithm, and zero target-family split overlap.
- It regenerates the source inventory, 356-entry data dictionary, aggregate metrics, paired statistics, interaction tables, overlap tables, and all 26 PNG figures from the stored CSV artifacts.
- The PDF is compiled twice so table-of-contents entries, references, and float numbering resolve before delivery.

Requested analysis family	Status	Evidence or reason
Performance summary and ranking	Complete	Ten retrieval systems, four synthetic algorithms, all denominators, scopes, and categories.
Paired significance and effects	Complete	Bootstrap intervals, exact McNemar tests, gains/losses, matched odds ratios.
Training dynamics	Unsupported	Algorithms are rule-based retrieval systems; no epochs, losses, gradients, seeds, or learning-rate logs exist.
Controlled comparisons	Complete	One-component Algorithm 4 ablations and eight equal-distance policies.
Distribution and error analysis	Complete	Distance, operations, evidence, length, cohorts, models, targets, categories, and worst examples.
Calibration and confusion matrices	Unsupported	Search outputs are ranked families without calibrated class probabilities or one fixed classification label space.
Ablation and sensitivity	Complete	Twenty-three component removals plus complete Algorithm 4; development and holdout tie sensitivity.
Correlation and interactions	Complete	Spearman matrix and edit-distance by bigram interaction heatmap.
Compute efficiency	Partial	Warm latency, preparation time, latency distributions, Pareto frontier, and relative radar profile; no CPU, GPU, memory, or energy logs.
Human outcomes	Unsupported	Balanced assignments exist, participant responses do not.
Leakage and split checks	Complete	Zero verified-family overlap between development and holdout; all synthetic source rows align across algorithms.

Table 12: Unsupported analyses are omitted rather than filled with mock values or inferred outcomes. The report generator fails if key row counts or split contracts do not reconcile.